

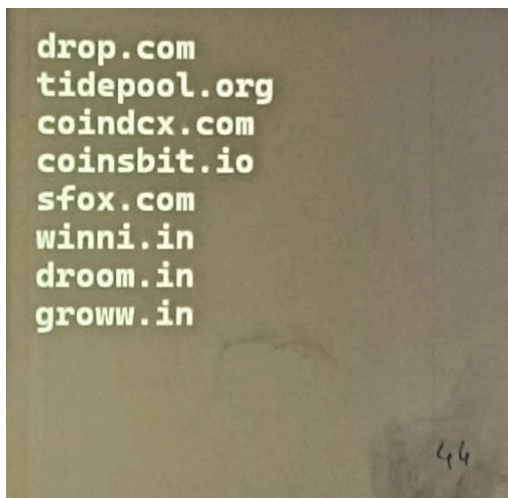
NMAP REPORT

Perform a comprehensive reconnaissance and service enumeration of the target host using Nmap. The scan should identify all open TCP ports, detect running services and their versions, determine the target operating system, execute default and safe NSE scripts, perform traceroute, and identify any potential vulnerabilities. Save the scan results in XML format, convert them into a professional HTML report, and generate a PDF version of the report. The final report should clearly explain each detected attribute, including host information, port states, services, versions, OS details, NSE script findings, traceroute information, and any identified security issues or recommendations.

Or, in a concise form:

Using Nmap, perform a comprehensive security assessment of the target host by identifying all open ports, running services, service versions, operating system, traceroute, and potential vulnerabilities. Generate a detailed HTML and PDF report explaining each scan result and its security significance.

Target ip : 192.168.74.130 (metasploitable 2)



```
sudo nmap \  
-sS \  
-sV \  
-sC \  
-O \  
-A \  
-p- \  
--version-all \  
--reason \  
--traceroute \  
--script "default,vuln,safe" \  
-T4 \  
-oA full_report \  
192.168.74.130
```

This generates:

```
full_report.nmap  
full_report.xml  
full_report.gnmap
```

```
(root@kali)-[~]  
# sudo nmap \  
-sS \  
-sV \  
-sC \  
-O \  
-A \  
-p- \  
--version-all \  
--reason \  
--traceroute \  
--script "default,vuln,safe" \  
-T4 \  
-oA full_report \  
192.168.74.130  
  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-05 04:09 -0400  
No profinet devices in the subnet  
Pre-scan script results:  
| broadcast-igmp-discovery:  
|   192.168.74.1  
|     Interface: eth0  
|     Version: 2  
|     Group: 224.0.0.252  
|     Description: Link-local Multicast Name Resolution (rfc4795)  
|   192.168.74.135  
|     Interface: eth0  
|     Version: 2  
|     Group: 224.0.0.251  
|     Description: mDNS (rfc6762)  
|   192.168.74.1  
|     Interface: eth0  
|     Version: 2
```

Explanation of Every Option

Option	Purpose	Report Information
<code>-sS</code>	TCP SYN Scan	Detects open TCP ports efficiently.
<code>-sV</code>	Service Version Detection	Identifies service names and versions (e.g., Apache 2.4.58, OpenSSH 9.6).
<code>-sC</code>	Default NSE Scripts	Runs common security and information-gathering scripts.
<code>-O</code>	OS Detection	Estimates operating system and kernel version.
<code>-A</code>	Aggressive Scan	Enables OS detection, version detection, NSE scripts, and traceroute.
<code>-p-</code>	Scan all 65,535 TCP ports	Finds services running on non-standard ports.
<code>--version-all</code>	Deep version detection	Uses all probes for better service fingerprinting.
<code>--reason</code>	Explain why a port is in its state	Shows evidence like SYN-ACK or RST.
<code>--traceroute</code>	Network path discovery	Displays hops between scanner and target.
<code>--script "default,vuln,safe"</code>	NSE scripts	Checks for common vulnerabilities and gathers additional information.
<code>-T4</code>	Faster timing	Speeds up scanning on reliable networks.
<code>-oA full_report</code>	Save in all formats	Creates <code>.nmap</code> , <code>.xml</code> , and <code>.gnmap</code> outputs.

Here's a mapping between the **Nmap options (attributes)** in your command and **which part of the report they produce**.

Nmap Command

```
sudo nmap \  
-sS \  
-sV \  
-sC \  
-O \  
-A \  
-p- \  
--version-all \  
--reason \  
--traceroute \  
--script "default,vuln,safe" \  
-T4 \  
-oA full_report \  
192.168.74.130
```

Nmap Option	Report Section Generated	Example Output
(default host discovery)	Host Status	Host is up
(default host discovery)	Latency	0.00097s latency
-sS	Port Scan	21/tcp open, 22/tcp open
-p-	All TCP Ports	Scans ports 1-65535 instead of only the top 1000
--reason	Port State Reason	syn-ack ttl 64
-sV	Service Detection	ftp, ssh, http
-sV	Product Detection	vsftpd, OpenSSH, Apache
-sV	Version Detection	2.3.4, 4.7p1
--version-all	Deep Version Detection	Improves software/version identification
-sC	Default NSE Scripts	ftp-anon, ftp-syst, ssh-hostkey, http-title
--script default	Additional Enumeration	Banner grabbing, titles, FTP info, SMB info, etc.
--script vuln	Vulnerability Detection	CVE-2011-2523, exploit checks
--script vuln	CVE Listing	CVE IDs, exploit references, CVSS scores
--script safe	Safe Information Gathering	SSL info, HTTP headers, DNS info, FTP details
-O	OS Detection	Linux, Windows, Kernel version, Device type
-O	TCP/IP Fingerprint	Used internally for OS detection
--traceroute	Network Path	Router hops to target

-A	Enables Multiple Features	Equivalent to <code>-O + -sV + -sC + --traceroute</code>
-T4	Scan Speed	Faster scan (doesn't add report content)
-oA full_report	Output Files	<code>full_report.nmap, .xml, .gnmap</code>

Which Option Generated Each Section?

Report Section	Nmap Option
Host Status	Default Host Discovery
Latency	Default Host Discovery
Open Ports	<code>-sS</code>
Port State	<code>-sS</code>
Reason (syn-ack ttl 64)	<code>--reason</code>
Service Name	<code>-sV</code>
Product Name	<code>-sV</code>
Software Version	<code>-sV, --version-all</code>
Banner	<code>-sC / default</code> scripts
Anonymous FTP Login	<code>ftp-anon (-sC)</code>
FTP Status	<code>ftp-syst (-sC)</code>
SSH Host Keys	<code>ssh-hostkey (-sC)</code>
HTTP Title	<code>http-title (-sC)</code>
SSL Certificate	<code>ssl-cert (-sC or safe)</code>
Vulnerabilities	<code>--script vuln</code>
CVE IDs	<code>--script vuln</code>
CVSS Score	<code>vulners</code> NSE script

Exploit References	<code>vulners</code> NSE script
OS Detection	<code>-O</code>
Device Type	<code>-0</code>
Network Distance	<code>-0</code>
Traceroute	<code>--traceroute</code>
Scan Statistics	Nmap (automatic)

Important Note about `-A`

The `-A` (Aggressive Scan) option is a **shortcut** that automatically enables:

```
-A
├── -O          → OS Detection
├── -sV         → Version Detection
├── -sC         → Default NSE Scripts
└── --traceroute → Traceroute
```

So, if you use `-A`, you technically don't need to specify `-O`, `-sV`, `-sC`, or `--traceroute` separately unless you want to make the command explicit for learning or documentation.

```
(root@kali)-[~]
# ls
full_report.gnmap  full_report.nmap  full_report.xml  nmap.sh
```

Method 1 (Recommended)

Install `xsltproc` if it isn't already installed:

```
sudo apt update
```

```
sudo apt install xsltproc -y
```

Convert the XML to HTML:

```
xsltproc /usr/share/nmap/nmap.xsl full_report.xml -o full_report.html
```

Method 2 (Using Nmap's XML stylesheet)

If your XML already contains the stylesheet reference, simply run:

```
xsltproc full_report.xml -o full_report.html
```

```
(root@kali)-[~]
# sudo apt update
sudo apt install xsltproc -y
Get:1 http://kali.download/kali kali-rolling InRelease [34.0 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 Packages [21.3 MB]
Get:3 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [53.5 MB]
Get:4 http://kali.download/kali kali-rolling/contrib amd64 Packages [103 kB]
Get:5 http://kali.download/kali kali-rolling/contrib amd64 Contents (deb) [188 kB]
Get:6 http://kali.download/kali kali-rolling/non-free amd64 Packages [169 kB]
Get:7 http://kali.download/kali kali-rolling/non-free amd64 Contents (deb) [887 kB]
Get:8 http://kali.download/kali kali-rolling/non-free-firmware amd64 Packages [15.4 kB]
Get:9 http://kali.download/kali kali-rolling/non-free-firmware amd64 Contents (deb) [40.1 kB]
Fetched 76.2 MB in 11s (7,181 kB/s)
467 packages can be upgraded. Run 'apt list --upgradable' to see them.
xsltproc is already the newest version (1.1.45-0.1).
xsltproc set to manually installed.
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 467
(root@kali)-[~]
# xsltproc /usr/share/nmap/nmap.xsl full_report.xml -o full_report.html
```

```
(root@kali)-[~]
# ls
full_report.gnmap  full_report.html  full_report.nmap  full_report.xml  nmap.sh  pass.txt  reson
```

```
(root@kali)-[~]
# ip -br a
lo                UNKNOWN    127.0.0.1/8 ::1/128
eth0              UP         192.168.74.129/24 fe80::1b6d:c5b0:2cd4:70ba/64

(root@kali)-[~]
# python3 -m http.server 8787
Serving HTTP on 0.0.0.0 port 8787 (http://0.0.0.0:8787/) ...
192.168.74.1 - - [05/Jul/2026 06:11:53] "GET / HTTP/1.1" 200 -
192.168.74.1 - - [05/Jul/2026 06:11:53] "GET / HTTP/1.1" 200 -
192.168.74.1 - - [05/Jul/2026 06:11:53] code 404, message File not found
192.168.74.1 - - [05/Jul/2026 06:11:53] "GET /favicon.ico HTTP/1.1" 404 -
192.168.74.1 - - [05/Jul/2026 06:11:57] "GET /full_report.html HTTP/1.1" 200 -
```


Search in browser

←→↻

Not secure192.168.74.129:8787/full_report.html

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Nmap Scan Report - Scanned at Sun Jul 5 04:09:21 2026

Scan Summary | Pre-Scan Script Output | 192.168.74.130 | Post-Scan Script Output

Scan Summary

Nmap 7.99 was initiated at Sun Jul 5 04:09:21 2026 with these arguments:
/usr/lib/nmap/nmap -ss -sC -O -A -p --reason-all --reason --traceroute --script default,vuln,safe -T4 -oA full_report 192.168.74.130
Verbosity: 0; Debug level 0
Nmap done at Sun Jul 5 04:22:52 2026; 1 IP address (1 host up) scanned in 810.32 seconds

Pre-Scan Script Output

Script Name	Output
broadcast-igmp-discovery	192.168.74.1 Interface: eth0 Version: 2 Group: 224.0.0.252 Description: Link-local Multicast Name Resolution (rfc4795) 192.168.74.135 Interface: eth0 Version: 2 Group: 224.0.0.251 Description: mDNS (rfc6762) 192.168.74.1 Interface: eth0 Version: 2 Group: 239.255.182.18 Description: Organization-Local Scope (rfc2365) 192.168.74.1 Interface: eth0 Version: 2 Group: 239.255.255.250 Description: Organization-Local Scope (rfc2365) Use the newtargets script-arg to add the results as targets
targets-sniffer	Sniffed 20 address(es). 224.0.0.1 224.0.0.9 239.255.255.253 224.0.0.251 239.255.255.250 192.168.74.254 192.168.74.133 0.0.0.0 192.168.74.132 192.168.74.1 239.255.182.18 224.0.0.252

ctrl+p

Nmap Scan Report - Scanned at Sun Jul 5 04:09:21 2026

Scan Summary

Nmap 7.99 was initiated at Sun Jul 5 04:09:21 2026 with these arguments:
/usr/lib/nmap/nmap -ss -sC -O -A -p --reason-all --reason --traceroute --script default,vuln,safe -T4 -oA full_report 192.168.74.130
Verbosity: 0; Debug level 0
Nmap done at Sun Jul 5 04:22:52 2026; 1 IP address (1 host up) scanned in 810.32 seconds

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targets-sniffer	Sniffed 20 address(es). 224.0.0.1 224.0.0.9 239.255.255.253 224.0.0.251 239.255.255.250 192.168.74.254 192.168.74.133 0.0.0.0 192.168.74.132 192.168.74.1 239.255.182.18 224.0.0.252 FAddr: 192B:128F:F461:80A FAddr: 192B:128F:F461:80A FAddr: 192B:128F:F461:80A 192.168.74.1 224.0.0.251 224.0.0.252 192.168.74.135 192.168.74.133
broadcast-ping	IP: 192.168.74.2 MAC: 08:00:56:FE:4E:16 Use --script-arg=newtargets to add the results as targets
Hosts:rdnss-rdntls	*TEMPORARILY DISABLED* due to changes in Rdnss's API. See https://www.rdnss.com/api/
Ndp-rdnss-rtt-and-ns	*TEMPORARILY DISABLED* due to changes in Rdnss's API. See https://www.rdnss.com/api/
broadcast-cgmp-info	
broadcast-flooding	eth0 ARP Request sender ip: 192.168.74.2 sender mac: 08:00:56:FE:4E:16 target ip: 192.168.74.133 eth0 ICMP seq id: 111 ip: 192.168.74.254 192.168.74.133 239.255.255.0 192.168.74.2 192.168.74.2 192.168.74.254 192.168.74.132 239.255.255.0 192.168.74.2 192.168.74.2
targets-scan	targets-scan is a mandatory parameter
broadcast-ethip-discover	Response 1 of 1: IP Interface: eth0 192.168.74.133 Server Identifier: 192.168.74.254 Subnet Mask: 255.255.255.0 Router: 192.168.74.2 Default Name Server: 192.168.74.2 Domain Name: localdomain Broadcast Address: 192.168.74.255 NetBios Name Server: 192.168.74.2
targets-ipv6-multicast-echo	IP: F46B:128F:F461:80A MAC: 08:00:56:FE:4E:16 IFA6: eth0 IP: F46B:128F:F461:80A MAC: 08:00:56:FE:4E:16 IFA6: eth0 Use --script-arg=newtargets to add the results as targets
targets-ipv6-multicast-invalid-dst	IP: F46B:128F:F461:80A MAC: 08:00:56:FE:4E:16 IFA6: eth0 IP: F46B:128F:F461:80A MAC: 08:00:56:FE:4E:16 IFA6: eth0 Use --script-arg=newtargets to add the results as targets

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Nmap Scan Report - Scanned at Sun Jul 5 04:09:21 2026

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1 of 58

Nmap Scan Report - Scanned at Sun Jul 5 04:09:21 2026

Scan Summary

Nmap 7.99 was initiated at Sun Jul 5 04:09:21 2026 with these arguments:
/usr/bin/nmap -sC -sV -sC -O -A -p --version-all --reason --script default,vuln,safe -T4 -oA full_report 192.168.74.130
Verbosity: 0; Debug level 0
Nmap done at Sun Jul 5 04:22:52 2026; 1 IP address (1 host up) scanned in 810.32 seconds

Pre-Scan Script Output

Script Name	Output
	192.168.74.1 Interface: eth0 Version: 2 Group: 224.0.0.252 Description: Link-local Multicast Name Resolution (rfc4795) 192.168.74.135 Interface: eth0 Version: 2 Group: 224.0.0.251 Description: mDNS (rfc6762) 192.168.74.1 Interface: eth0 Version: 2 Group: 239.255.102.18 Description: Organization-Local Scope (rfc2365) 192.168.74.1 Interface: eth0 Version: 2 Group: 239.255.255.250 Description: Organization-Local Scope (rfc2365) Use the newtargets script-arg to add the results as targets
broadcast-igmp-discovery	
	Sniffed 20 address(es). 224.0.0.1 224.0.0.9 239.255.255.253 224.0.0.251 239.255.255.250 192.168.74.254 192.168.74.133 0.0.0.0 192.168.74.132 192.168.74.1 239.255.102.18 224.0.0.252 fe80::1b6d:c5b0:2cda:70ba fe80::28c:29ff:fe91:8cb fe80::28c:29ff:fe91:8cb 192.168.74.2 224.0.0.13 224.0.23.12 192.168.74.135 192.40.210.39
targets-sniffer	
	IP: 192.168.74.2 MAC: 08:59:56:fb:a6:be Use --script-args=newtargets to add the results as targets
broadcast-ping	
hostmap-robotex	*TEMPORARILY DISABLED* due to changes in Robotex's API. See https://www.robotex.com/api/
http-robotex-shared-ns	*TEMPORARILY DISABLED* due to changes in Robotex's API. See https://www.robotex.com/api/
broadcast-upnp-info	